

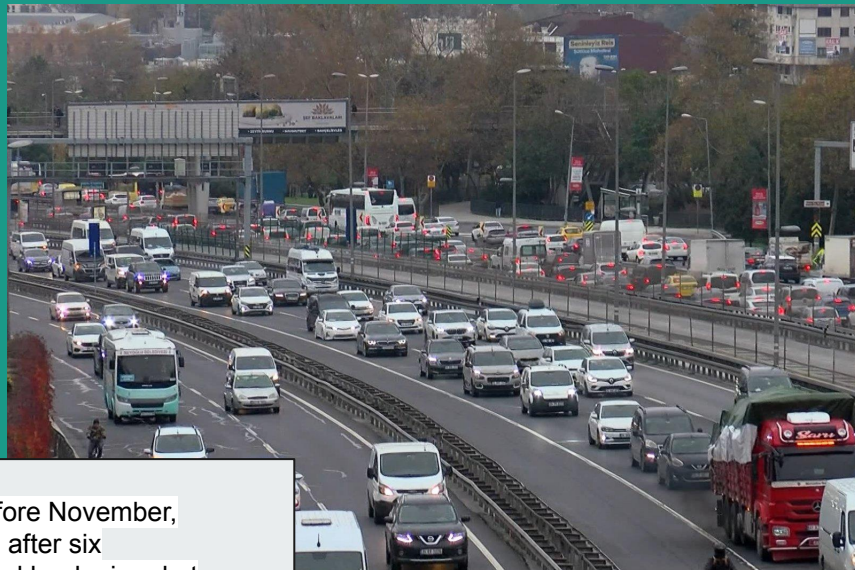
Car Market Analysis: Investment Strategy in Turkiye

Analyzing Car Investments in a High Inflation Economy



Turkiye: Newspaper articles

Car registrations are increasing



Sales registered 10 consecutive monthly records before November, despite a sharp rise in the costs of loans for vehicles after six successive months of interest rate hikes by the central bank aimed at cooling demand and stemming inflation.

2023-12-20 [Link](#)

“Some are even buying a third or a fourth vehicle,”

Alpagut Girgin, general manager of Opel [Turkey](#)
2022 June [Link](#)

Car owners generally change their vehicles every four to five years, but nowadays, people are trying to sell even their three-year old cars:

“[...] they think the price of the car they want to buy will rise in the near future.”

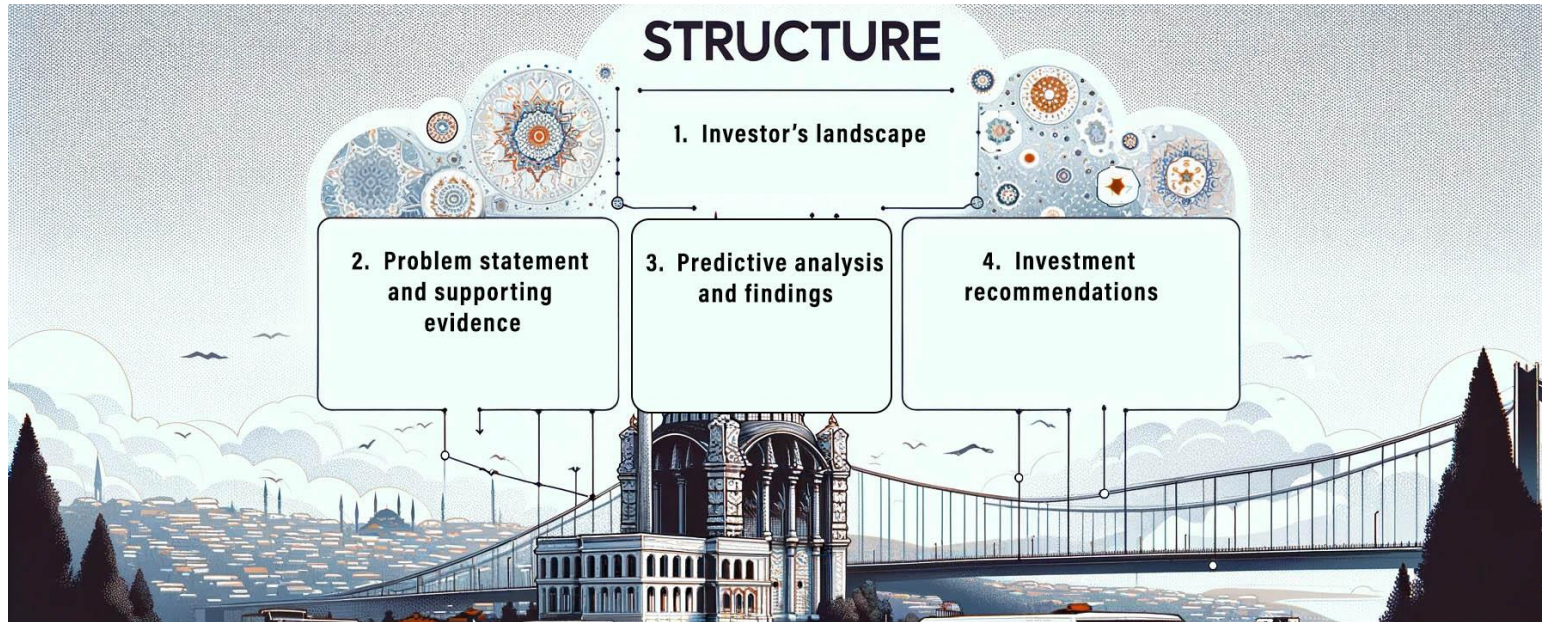


Project Mission

Provide used car investment advice to a fictitious Turkish investor, looking to invest the equivalent of \$100,000 USD for a 3-year period

Presentation Claim and Structure

Claim: Investing in cars is seen as a hedge against inflation in Turkey.



1.0 Investor's Landscape

Turkiye: Macroeconomic Context

Yearly Inflation
(2022) 72.3%
(2023) 53.86%

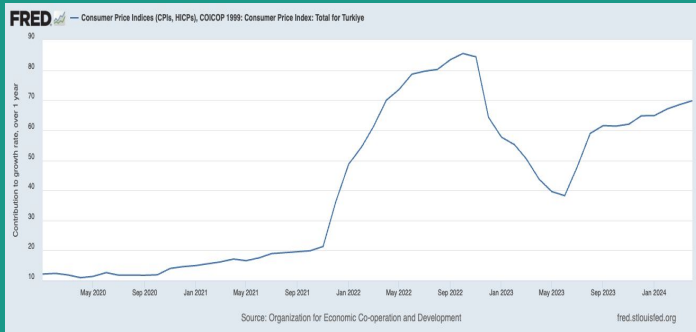
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1 USD:
5.86 TRY (2020-01-16)
32.22 TRY (2024-06-06)

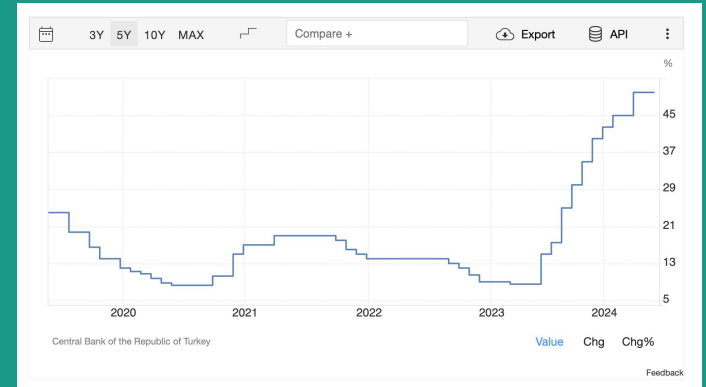
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Bank Interest Rate
8.25% (2020-07)
50% (since 2024-03)

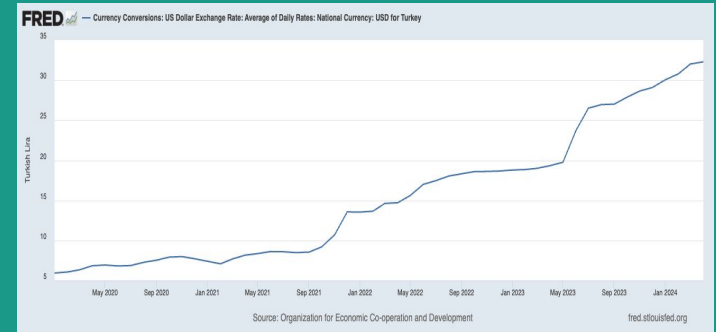
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Consumer Price Index: Total for Türkiye 2020-01-01 - 2024-04-01
Monthly, Contribution to growth rate, over 1 year,



Central Bank of the Republic of Turkey: Interest Rate



Currency Conversions: US Dollar Exchange Rate:
Average of Daily Rates: National Currency: USD for Turkey

Turkiye: Household Car Budget

Distribution of **annual** equivalised household disposable income by quintiles ordered by equivalised household disposable income, 2022, 2023

[TR Türkiye, SR Level 2]

	Quintiles											
	Total		First quintile		Second quintile		Third quintile		Fourth quintile		Last quintile	
	2022	2023	2022	2023	2022	2023	2022	2023	2022	2023	2022	2023
TR Türkiye												
Percentage (%)	100.0	100.0	6.0	5.9	10.4	9.8	14.7	14.0	20.9	20.5	48.0	49.8
Mean (TL)	48 642	83 808	14 712	24 780	25 304	41 196	35 652	58 726	50 781	85 717	116 773	208 653
Median (TL)	35 493	58 374	15 393	25 887	25 161	41 147	35 493	58 382	50 241	84 676	84 559	148 800
TR10 (İstanbul)												
Percentage (%)	100.0	100.0	6.4	6.1	9.9	9.6	13.5	13.6	18.6	19.1	51.6	51.6
Mean (TL)	69 904	114 634	22 278	35 069	34 744	55 481	47 198	77 880	64 993	109 101	180 379	295 856
Median (TL)	47 204	77 508	23 176	35 786	34 816	55 429	47 253	77 508	64 017	107 800	113 744	191 620
Average Exchange Rate (USD - TRY)	16.6	23.8	16.6	23.8	16.6	23.8	16.6	23.8	16.6	23.8	16.6	23.8
TR Türkiye												
Mean (TL)	\$2,934.58	\$3,519.35	\$887.56	\$1,040.59	\$1,526.58	\$1,729.94	\$2,150.91	\$2,466.07	\$3,063.65	\$3,599.51	\$7,044.93	\$8,761.93
Median (TL)	\$2,141.33	\$2,451.29	\$928.69	\$1,087.08	\$1,517.96	\$1,727.86	\$2,141.33	\$2,451.62	\$3,031.03	\$3,555.77	\$5,101.50	\$6,248.53
TR10 (İstanbul)												
Mean (TL)	\$4,217.34	\$4,813.79	\$1,344.03	\$1,472.64	\$2,096.11	\$2,329.80	\$2,847.48	\$3,270.38	\$3,921.03	\$4,581.46	\$10,882.33	\$12,423.82
Median (TL)	\$2,847.83	\$3,254.77	\$1,398.22	\$1,502.75	\$2,100.47	\$2,327.60	\$2,850.77	\$3,254.77	\$3,862.17	\$4,526.83	\$6,862.21	\$8,046.64

2023 annual total disposable income:

(3rd quintile, median, in Türkiye) ~\$2,500

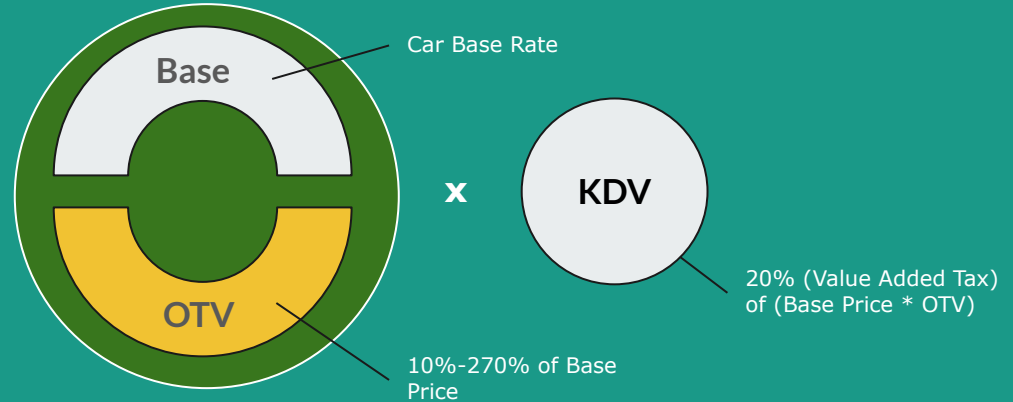
let's assume 30% of this can go toward car payments:

\$833.33 per year

*even the highest tier (Istanbul, Mean, Tier 5) would have \$3,727.15 to spend on a car per year

Formula to calculate a new car's price

Turkiye: Car Luxury Tax (ÖTV)



	Engine Size	CO2 Emissions (g/km)	Luxury Features	Petrol (ÖTV Rate)	Diesel (ÖTV Rate)	Hybrid (ÖTV Rate)	Electric (ÖTV Rate)	
	Up to 1600 cc	<100	Standard	45%	50%	20%	10%	
	Up to 1600 cc	100-150	Mid-Level	55%	60%	25%	15%	
	Up to 1600 cc	>150	High-End	80%	90%	40%	20%	
	1601 cc to 2000 cc	<100	Standard	100%	110%	50%	20%	
	1601 cc to 2000 cc	100-150	Mid-Level	120%	130%	60%	30%	
	1601 cc to 2000 cc	>150	High-End	150%	170%	80%	40%	
	Above 2000 cc	<100	Standard	200%	220%	100%	50%	
	Above 2000 cc	100-150	Mid-Level	220%	240%	120%	60%	
	Above 2000 cc	>150	High-End	250%	270%	150%	70%	

OTV encourages: small engines (< 1600 cc), low CO2 pollution, electric cars, low level of "Luxury Features" (standard radio, textile upholstery, manual climate control , halogen lights)

Turkiye: Car prices comparison

- Out the door
- Lowest equipment tier
- Prices converted to USD



Fiat 500

US: \$18,850

TR: \$37,760

*even at 0% interest, 0% down, with
median disposable income of \$833.33/year:
45.3 years to pay off (10.1 yr highest tier)



Range Rover

US: \$101,400

TR: \$377,600

*even at 0% interest, 0% down, with
median disposable income of \$833.33/year:
453 years to pay off (101.3 yr highest tier)



Mercedes C-Class

US: \$48,200

TR: \$147,500

*even at 0% interest, 0% down, with
median disposable income of \$833.33/year:
177 years to pay off (39.6 yr highest tier)



VW Golf

US: \$27,860

TR: \$53,100

*even at 0% interest, 0% down, with
median disposable income of \$833.33/year:
63.7 years to pay off (14.24 yr highest tier)

Turkiye:

Against all odds

Car registrations are increasing



Sales: Retail Trade: Car Registration: Passenger Cars for Turkiye
2020-01-01 - 2024-04-01, Number, Monthly Level

Sketch

Even in Turkey's specific economic scenario, cars are not an investment per-se.

Someone who had 250,000 TL back in May 2021 (approx \$30K USD at the time)

- if they held on to the cash, it would now be worth approximately \$7.8K (74% loss).
- but if they had bought a car, they would still have approximately \$18.5K (38.4% loss)

Fiat Egea		Purchased in			
	USD - TL	May 2021	May 2022	May 2023	May 2024
May 2021	8.36	250,000 TL			
May 2022	15.61		475,000 TL		
May 2023	19.75			600,000 TL	
May 2024	32.21	600,000 TL	700,000 TL	800,000 TL	917,900 TL
		\$7,761.56			
Fiat Egea		Purchased in			
	USD - TL	May 2021	May 2022	May 2023	May 2024
May 2021	8.36	\$29,904.31			
May 2022	15.61		\$30,429.21		
May 2023	19.75			\$30,379.75	
May 2024	32.21	\$18,627.76	\$21,732.38	\$24,837.01	\$28,497.36
			Italy	€15,250.00	\$16,502.94
			Germany	€14,875.00	\$16,097.13
			France	€15,000.00	\$16,232.40



Project mission - course adjustment

We now understand that our investor will not make any absolute value.

People's rush to purchase cars comes from the high inflation, the idea that they bought the car with 250,000 TL and sold it with 600,000 TL, but in absolute value they still lost a significant amount.

Knowing this now, let's do our best to provide advice that will steer them towards minimizing their loss

Try to offer advice that beats a 38.4% loss over the next 3 years.

02. Data

Data Collection

Turkey Car Market 2020

Data Card Code (26) Discussion (0) Suggestions (0)

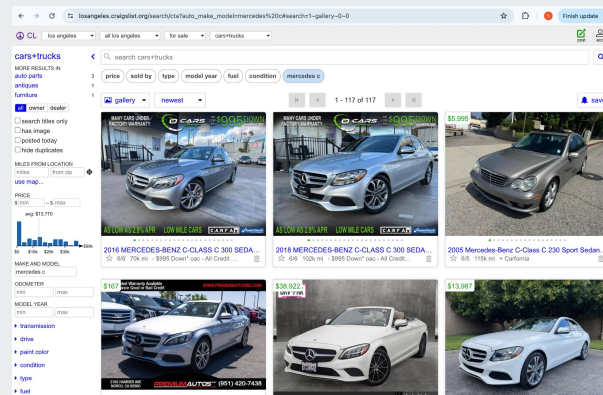
turkey_car_market.csv (1.26 MB)

Detail Compact Column 10 of 15 columns

About this file
This data includes Turkey vehicle prices

A İlan Tarihi Date	B Marka Brand	C Araç Tipi Grubu Brand Group	D Araç Tipi Vehicle Type	E Model Yılı Model Year	F Yakıt Fuel Type
18/04/2020	39% Renault	23% Clio	5% Diğer	11%	Dizel
17/04/2020	13% Fiat	7% Megane	5% 1.5 dCi Joy	5%	Benzin/LF
Other (4368)	48% Other (8311)	70% Other (8111)	90% Other (7602)	84%	Other (14)
27/05/2020	Jaguar	XF	2.0 D Prestige Plus	2017.0	Dizel
16/06/2020	Acura	CL	-	2015.0	Dizel
14/06/2020	Acura	CL	2.2	1994.0	Benzin/L
11/06/2020	Acura	CL	-	2013.0	Dizel
11/06/2020	Acura	CL	2.2	2010.0	Dizel
04/06/2020	Acura	CL	2.2	1999.0	Dizel
03/06/2020	Acura	CL	2.2	2014.0	Dizel

Kaggle - data scraped in 2020
(9,044 observations, 24 variables)



Craigslist - I scraped LA Craigslist for US used car data
(6,081 observations, 18 variables)

Arabam.com oto ekspertiz Uzman gözünden detaylı ekspertiz

Araç alım-satım süreçlerinizi oto ekspertiz adında da arabam.com güvencesi yanınızda!

Model	Yıl	Fiyat	Tarih	İlan Sahibi
Ford Fiesta 1.0 GTDI Titanium	2013	575,000 TL	06 Haziran 2024	Bahadır Barmis
Ford Transit Courier 1.5 TDCi Trend	2022	555,000 TL	06 Haziran 2024	Bursa Osmangazi
Ford Transit Courier 1.5 TDCi Trend	2023	575,000 TL	06 Haziran 2024	Bursa Osmangazi
Fiat Linea 1.3 Multijet Active Plus	2012	375,000 TL	06 Haziran 2024	İstanbul Etiler
Renault Megane 1.6 Joy	2019	815,000 TL	06 Haziran 2024	Ankara Çankaya

Arabam.com - I scraped a Turkish website for used car data
(72,674 observations, 24 variables)

Data Cleaning

Turkish dataframes

- translate columns to English
- translate dates to English and format as datetime64
- merge the two Turkish Data Sets
- use functions to clean Object data to be able to transform it to Int
- translate categorical values (color / owner / body type etc)

US dataframe

Import / clean / handle NaN / check for duplicates / keep Clean titles

```
column_rename_dict = {
    'İlan Tarihi': 'Ad Date',
    'Marka': 'Make',
    'Seri': 'Model',
    'Model': 'Package',
    'Yıl': 'Year',
    'Kilometre': 'Kilometers (km)',
    'Vites Tipi': 'Transmission',
    'Yakıt Tipi': 'Fuel Type',
    'Kasa Tipi': 'Body Type',
    'Renk': 'Color',
    'Motor Hacmi': 'Engine Size (cc)',
    'Motor Gücü': 'Engine Power (hp)',
    'Araç Durumu': 'Vehicle Condition',
    'Ort. Yakıt Tüketimi': 'Avg. Fuel Consumption (L/100km)',
    'Yakıt Deposu': 'Fuel Tank Size (L)',
    'Boya-değişen': 'Paint-changed',
    'Takasa Uygun': 'Willing to Exchange',
    'Kimden': 'From',
    'Fiyat': 'Price (Turkish Liras)',
    'Link': 'Link',
    'İlan No': 'Ad Number',
    'Page': 'Page',
    'Çekiş': 'Traction'
}
```

```
# Function to clean and convert 'Kilometers' column
def clean_kilometers(km_str):
    if isinstance(km_str, str):
        km_str = km_str.replace('.', '').replace(' km', '')
    if km_str != km_str: # Check if km_str is NaN
        return km_str # Return NaN as is
    return int(km_str)

# Function to clean and convert 'Avg. Fuel Consumption' and 'Fuel Tank'
def clean_fuel_data(fuel_str):
    if isinstance(fuel_str, str):
        fuel_str = fuel_str.replace(',', '.').replace(' lt', '')
    return float(fuel_str)

# Function to clean and convert 'Price' column
def clean_price(price_str):
    if isinstance(price_str, str):
        price_str = price_str.replace('.', '').replace(' TL', '')
    return int(price_str)
```



Data Cleaning

Standardize Make and Models

- Check value counts for current values and standardize the spelling of the Makes and Models
- This is so we can filter by make and model

```
# Standardize the "Make" column to have the
df_US['Make'] = df_US['Make'].str.capitalize
df_US['Model'] = df_US['Model'].str.capitalize
```

```
# Define a dictionary for the replacements
replacements = {
```

```
    'Chevorlet': 'Chevrolet',
    'Toyota': 'Toyota',
    'Hyundai': 'Hyundai',
    'Rolls': 'Rolls-Royce',
    'Rolls-royce': 'Rolls-Royce',
    'Rolls royce': 'Rolls-Royce',
    'Volkswagen': 'VW',
    'Vw': 'VW',
    'Bmw': 'BMW',
    'Ford,': 'Ford',
    'Mercedes-benz': 'Mercedes',
    'Mercedes': 'Mercedes',
    'Mercedesbenz': 'Mercedes',
    'Chevy': 'Chevrolet',
    'Range': 'Range Rover',
    'Range rover': 'Range Rover',
    'Landrover': 'Land Rover',
    'Land': 'Land Rover',
    'Land rover': 'Land Rover'
```

```
}
```

```
# Replace the values in the "Make" col
```

```
# Define a dictionary for model replacements
```

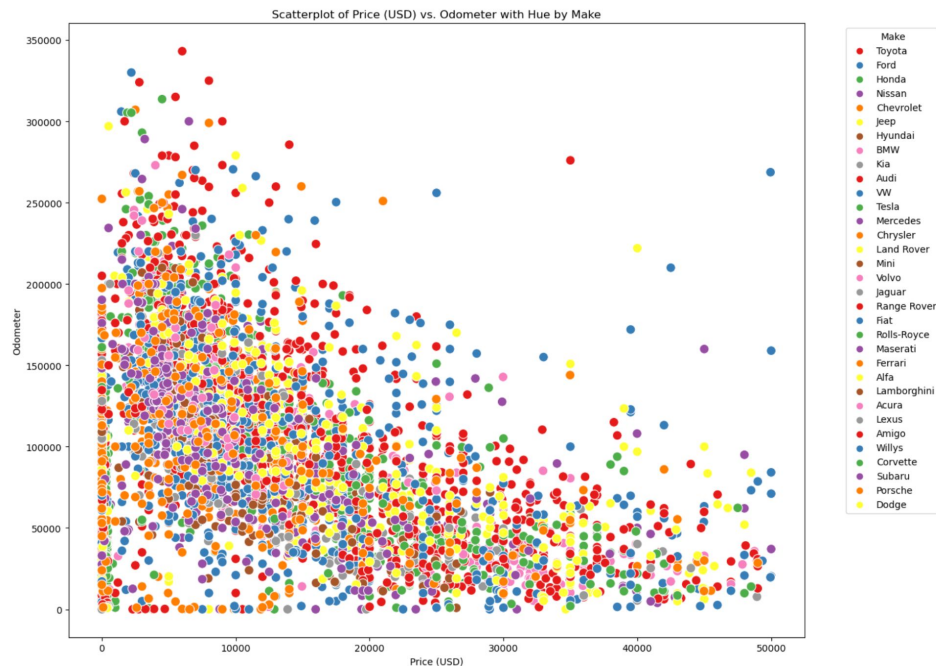
```
ford_model_replacements = {
    'F-150': 'F-150',
    'F150': 'F-150',
    'F-250': 'F-250',
    'F250': 'F-250',
    'F-350': 'F-350',
    'F350': 'F-350',
    'F-550': 'F-550',
    'F550': 'F-550',
    'Flex': 'Flex',
    'Mustang': 'Mustang',
    'Explorer': 'Explorer',
    'Escape': 'Escape',
    'Focus': 'Focus',
    'Fiesta': 'Fiesta',
    'Fusion': 'Fusion',
    'Ranger': 'Ranger',
    'Transit': 'Transit'
```

```
}
```

Exploratory Visualizations

US Data scatterplot of:
Price (USD) vs Odometer, Hue by Make

- To keep it comparable to Turkish cars, this data is filtered by:
< \$50,000 and
< 350,000 miles
- Despite a great variety of car brands we see a pretty uniform distribution where price is inversely correlated with mileage
- In the US cars have a pretty uniform distribution



Calculate USD Absolute (for Turkish Data)

- Import Turkish Inflation index and US Exchange rate since January 2020
- Use Turkish inflation rate to calculate historic amounts of TL and bring them to today's value; now we can compare any two cars from the Turkish dataframe
- Leave TL values from the past couple of months as they are since they are recent enough
- Use recent exchange rate (32.22 TL to 1 USD) to calculate everything in "USD Absolute". Now we are able to compare Turkish cars with US cars

```
# Initialize FRED
fred = Fred(api_key=fred_api_key)

# Fetch historical exchange rates (USD/TRY)
def get_exchange_rate_data(api_key, start_date='2020-01-01'):
    fred = Fred(api_key=api_key)
    exchange_rate = fred.get_series('CCUSMA02TRM618N', observation_start=start_date)
    exchange_rate_df = exchange_rate.reset_index()
    exchange_rate_df.columns = ['Date', 'Exchange Rate']
    return exchange_rate_df

# Fetch CPI data
def get_cpi_data(api_key, start_date='2020-01-01'):
    fred = Fred(api_key=api_key)
    cpi = fred.get_series('TURCPIALLMINMEI', observation_start=start_date)
    cpi_df = cpi.reset_index()
    cpi_df.columns = ['Date', 'CPI Index']
    return cpi_df

# Define start_date
start_date = '2020-01-01'

# Ensure the 'CPI Index' column is numeric
df_4yr['CPI Index'] = pd.to_numeric(df_4yr['CPI Index'], errors='coerce')

# Adjust prices based on CPI
df_4yr['Adjusted Price (Turkish Liras)'] = df_4yr.apply(
    lambda row: row['Price (Turkish Liras)'] if row['Ad Date Monthly'] == pd.Timestamp('2024-04-01')
    else (row['Price (Turkish Liras)'] / row['CPI Index'] * latest_cpi_index),
    axis=1
)

# Get the most recent exchange rate for April 2024
latest_exchange_rate = exchange_rate_df[exchange_rate_df['Date'] == '2024-04-01']['Exchange Rate Merged'].values[0]

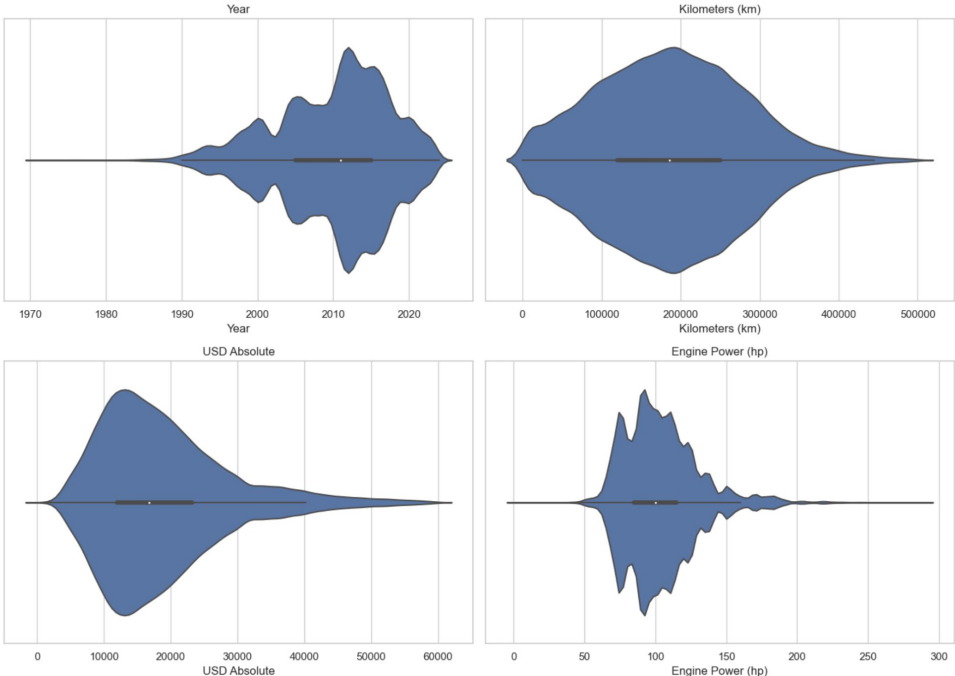
# Calculate USD Absolute as Adjusted Price (Turkish Liras) / the most recent exchange rate
df_4yr['USD Absolute'] = df_4yr['Adjusted Price (Turkish Liras)'] / latest_exchange_rate
```




Exploratory Visualizations

Violin plots for Turkish Data sets

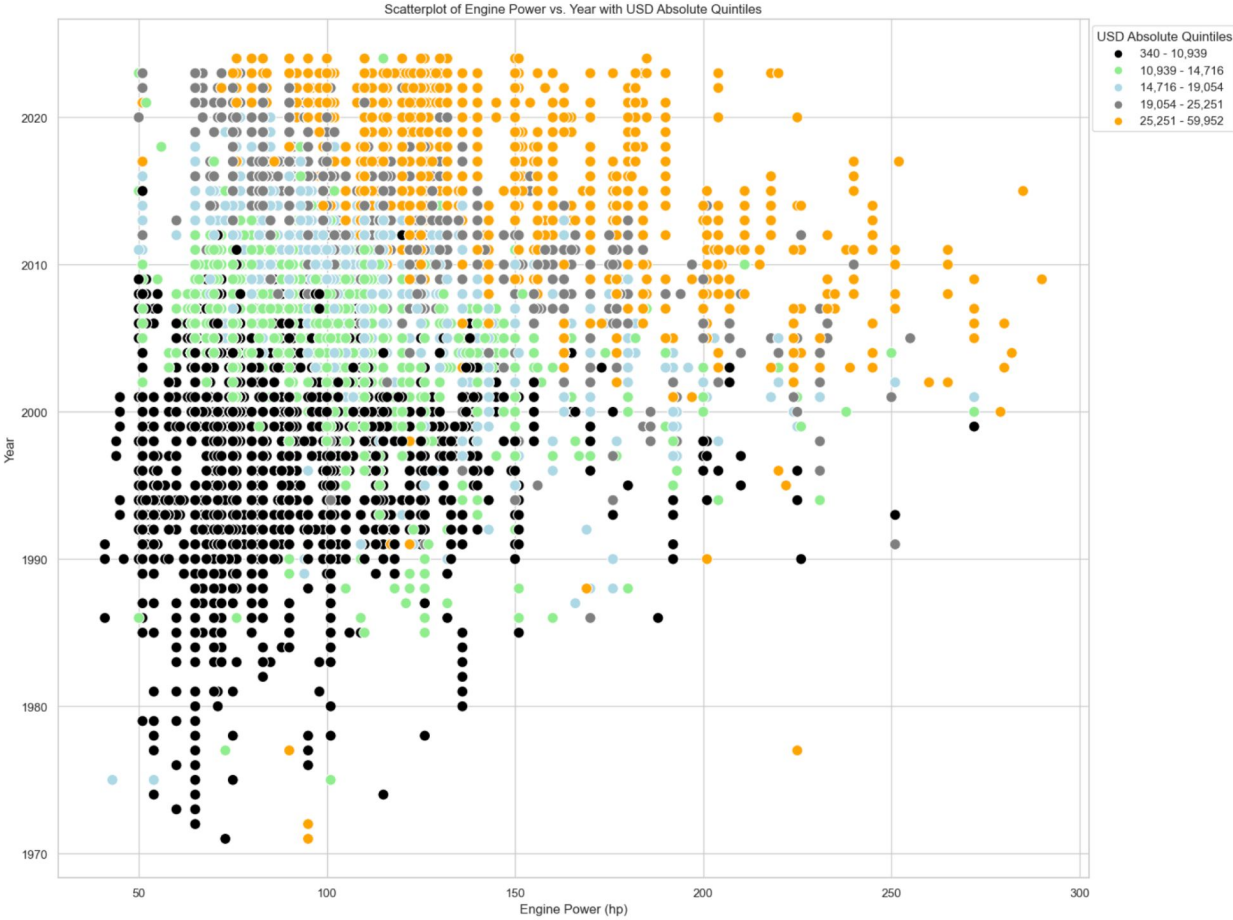
Most popular year: 2012
Most popular km: 200,000 km
Most popular price: \$12,000 USD
Most popular hp: 100 hp





Turkish data set:

- When plotting Engine Power vs Manufacture Year of the vehicles with a hue from quintiles, we can see a clear correlation of high price with high hp and new cars

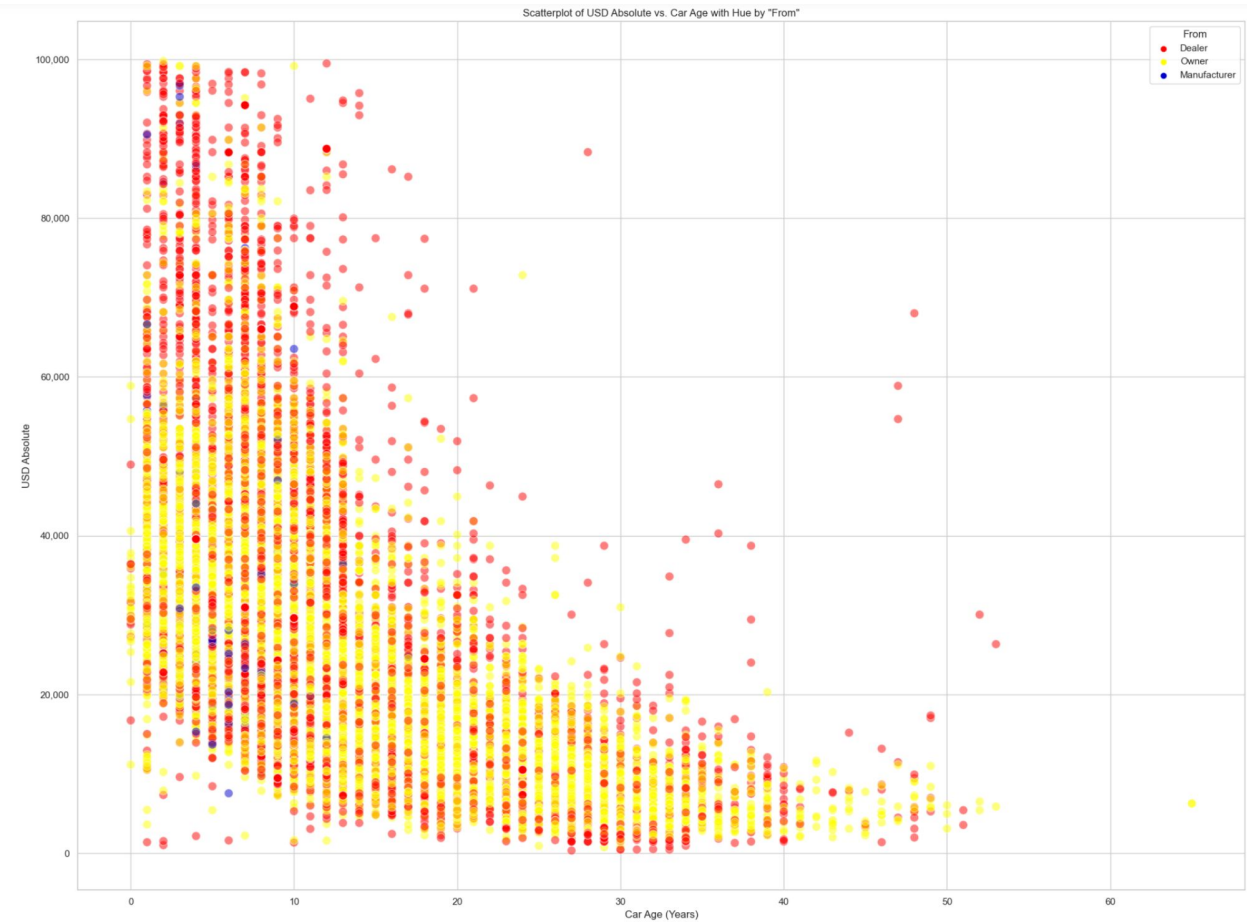


2.0 Data | Exploratory Visualizations



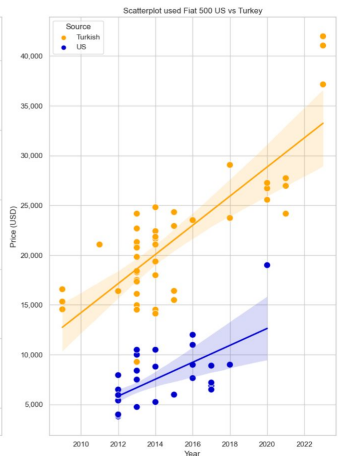
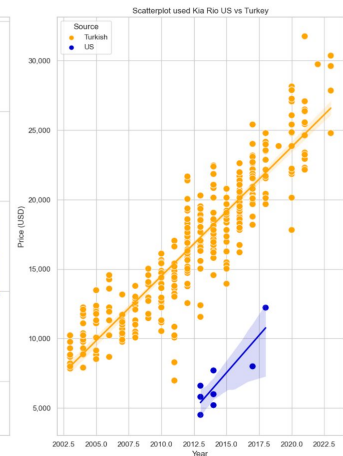
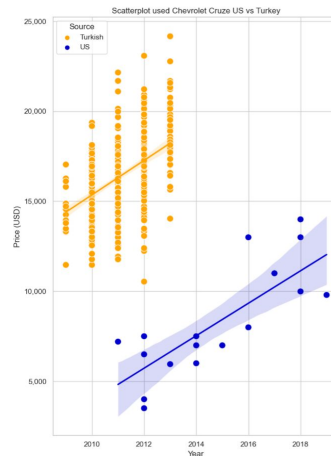
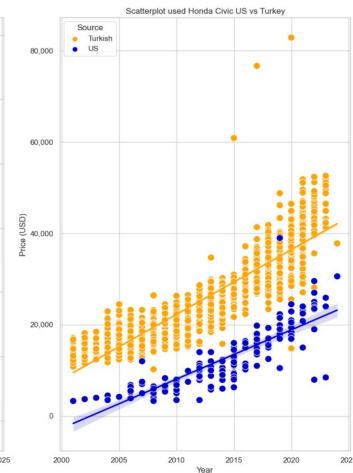
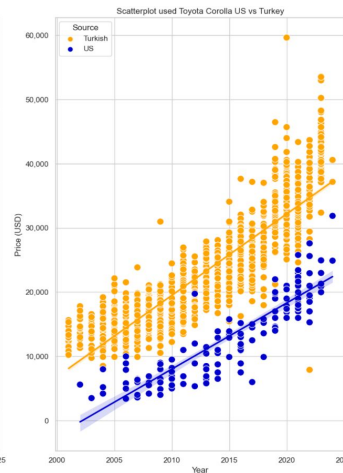
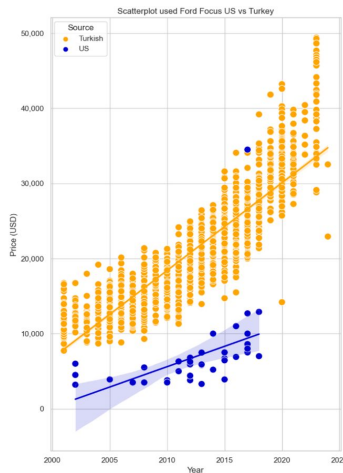
Turkish data set:

- Clear inverse correlation between price and manufacture year
- Cars sold by “Dealers” are more expensive than cars sold by “Owners”

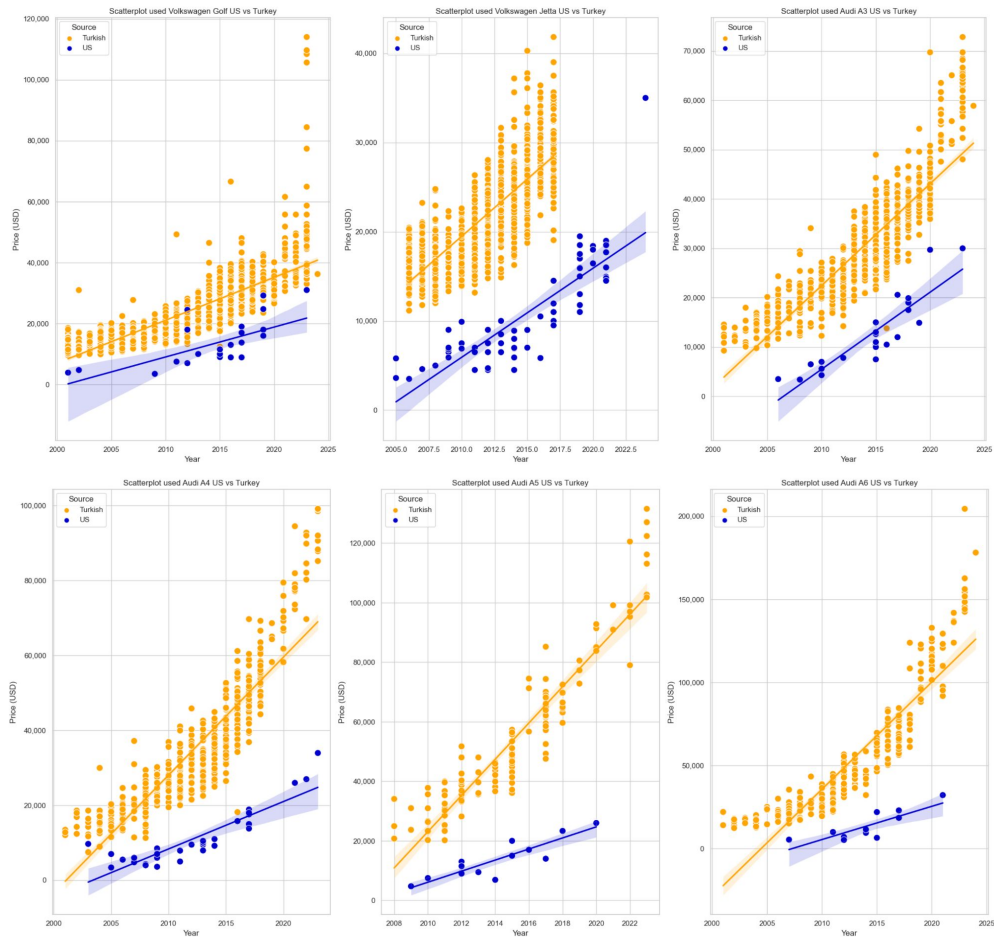


2.0 Data | Exploratory Visualizations

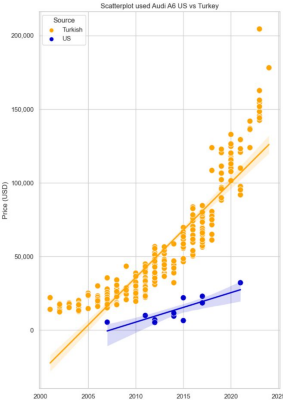
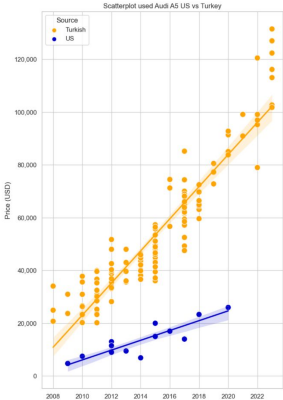
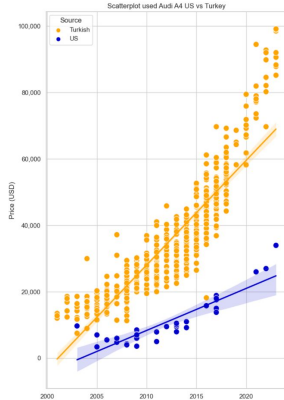
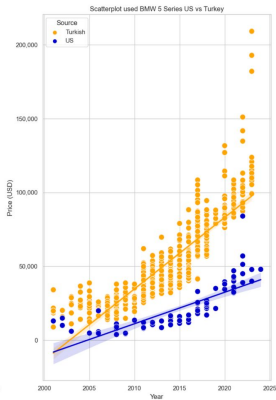
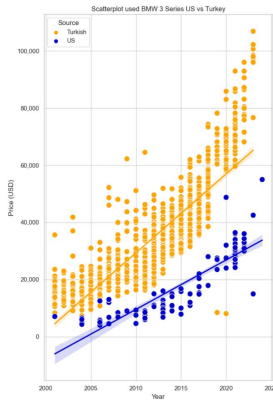
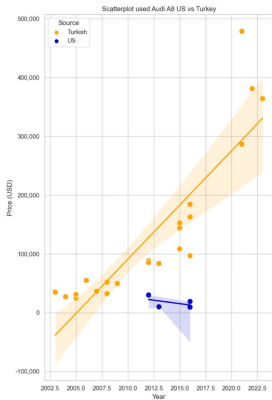
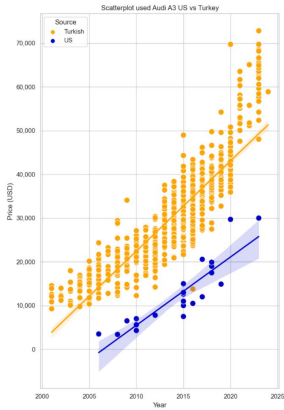
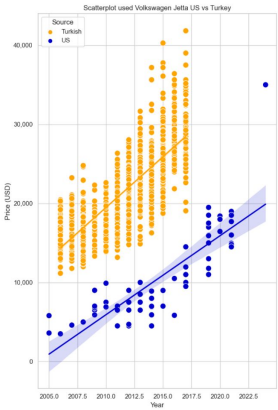
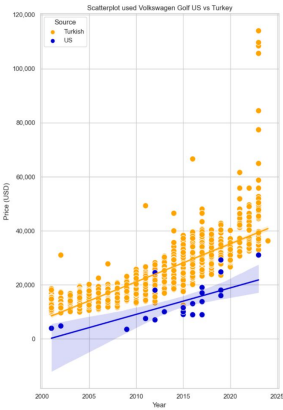
- Compare apples to apples between the Turkish and the US data sets
- In these and the following pages we can clearly see the group of US cars (blue) versus the Turkish group.
- Each scatterplot covers one make and model so when looking at a specific year we can see that the Turkish car is 2-3x more expensive than the same Make and Model in the US



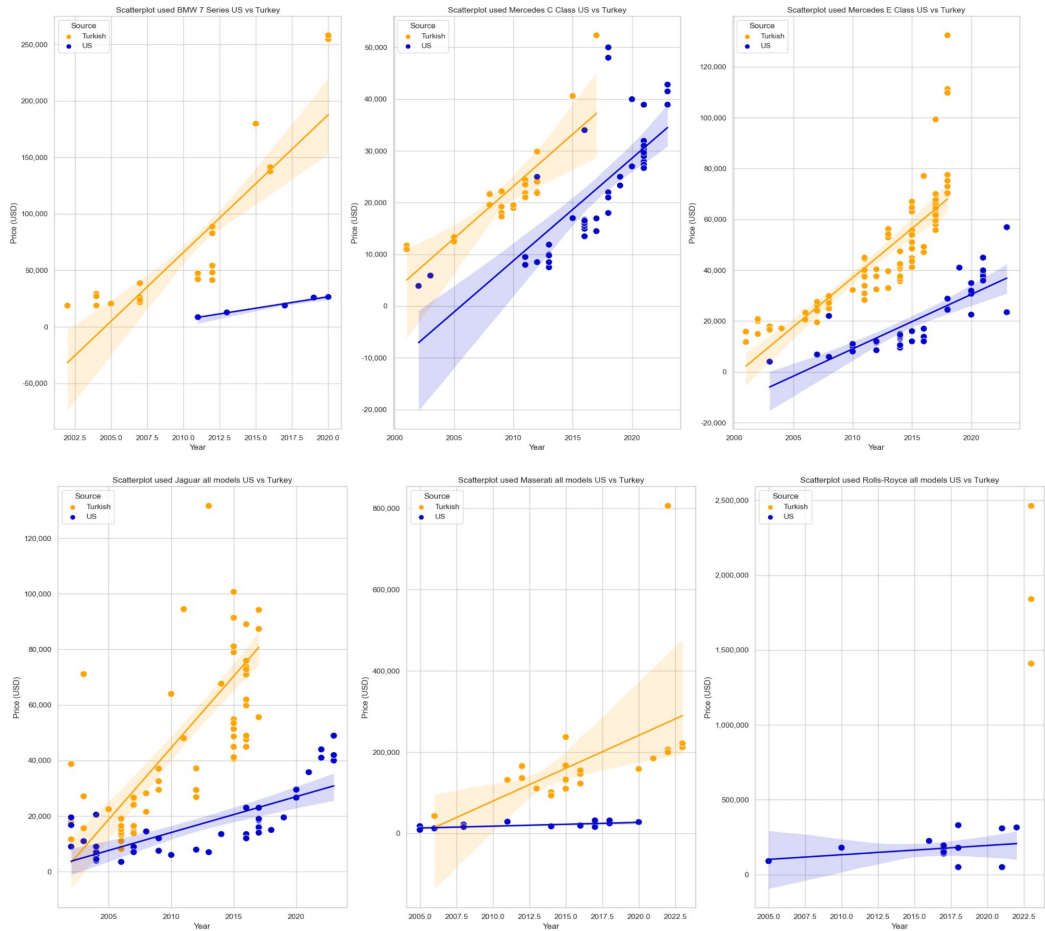
2.0 Data | Exploratory Visualizations



2.0 Data | Exploratory Visualizations



2.0 Data | Exploratory Visualizations

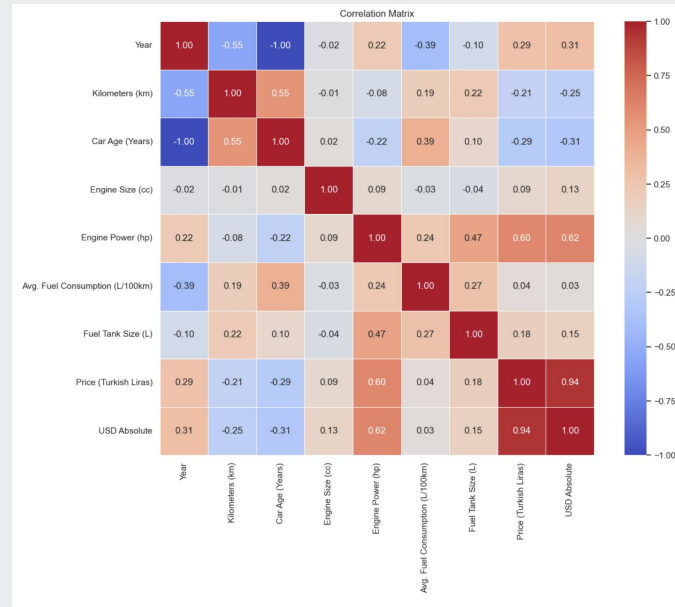


02. Predictive Analysis and findings

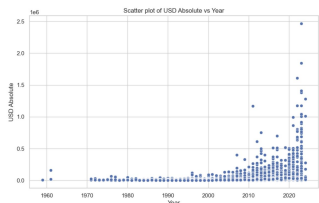
Descriptive Statistics

We will first try to run a linear regression

- Car Age (Years) and Price (Turkish Liras) will be removed



2.0 Data | Linear Regression



I noticed the quadratic relation between most values and USD so I decided to take log for “USD Absolute” (the dependent variable)

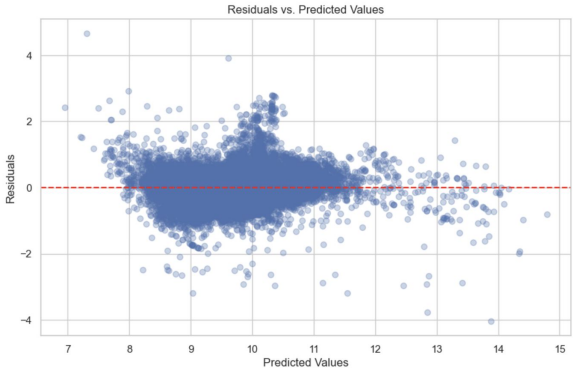
Very good results for the initial Linear Regression:

- Adj-R2 0.766
- Pretty low VIF values
- Ridge and Lasso cannot improve the R2

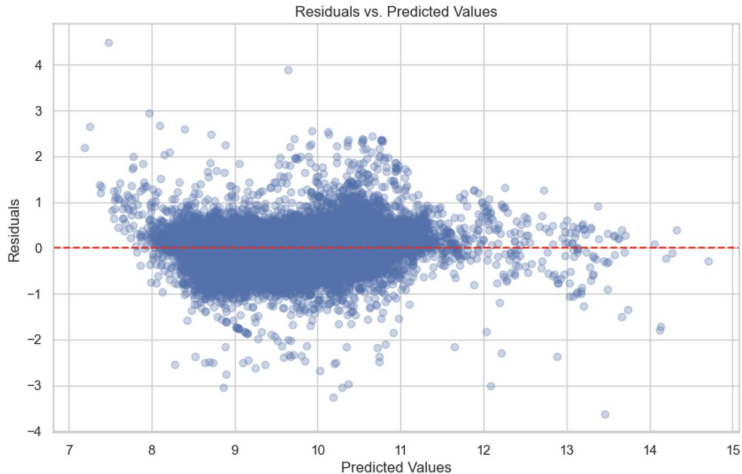
Results: Ordinary least squares					
Model:	OLS	Adj. R-squared:	0.766		
Dependent Variable:	log_USD_Absolute	AIC:	33417.3499		
Date:	2024-06-05 22:42	BIC:	33687.3445		
No. Observations:	81647	Log-Likelihood:	-16680.		
Df Model:	28	F-statistic:	9531.		
Df Residuals:	81618	Prob (F-statistic):	0.00		
R-squared:	0.766	Scale:	0.088130		

	Coef.	Std.Err.	t	P> t	[0.025	0.975]
const	9.7231	0.0010	9358.6410	0.0000	9.7211	9.7251
Year	0.3861	0.0013	288.2822	0.0000	0.3834	0.3887
Kilometers (km)	-0.0622	0.0013	-48.4135	0.0000	-0.0648	-0.0597
Engine Size (cc)	0.0189	0.0011	17.9863	0.0000	0.0169	0.0210
Engine Power (hp)	0.2149	0.0013	161.6384	0.0000	0.2123	0.2175
Fuel Tank Size (L)	0.0544	0.0013	43.3324	0.0000	0.0519	0.0568
Transmission_Manual	-0.0589	0.0014	-42.1981	0.0000	-0.0616	-0.0562
Transmission_Semi-Automatic	-0.0051	0.0014	-3.7994	0.0001	-0.0078	-0.0025
Fuel Type_Electric	-0.0095	0.0011	-9.0059	0.0000	-0.0115	-0.0074
Fuel Type_Gasoline	0.0232	0.0012	20.0210	0.0000	0.0209	0.0255
Fuel Type_Hybrid	0.0049	0.0011	4.6879	0.0000	0.0029	0.0070
Fuel Type_LPG and Gasoline	0.0119	0.0012	9.8386	0.0000	0.0095	0.0143
Body Type_Cabrio	0.0088	0.0010	8.4152	0.0000	0.0067	0.0108
Body Type_Coupe	0.0143	0.0011	13.5173	0.0000	0.0122	0.0163
Body Type_Crossover	-0.0043	0.0010	-4.1435	0.0000	-0.0064	-0.0023
Body Type_MPV	-0.0011	0.0010	-1.0629	0.2878	-0.0031	0.0009
Body Type_Other	-0.0084	0.0011	-7.8068	0.0000	-0.0105	-0.0063
Body Type_SUV	-0.0033	0.0010	-3.1925	0.0014	-0.0054	-0.0013
Body Type_Station Wagon	-0.0079	0.0010	-7.5743	0.0000	-0.0099	-0.0059
Body Type_Van	-0.0072	0.0010	-6.8405	0.0000	-0.0092	-0.0051
Vehicle Condition_New	0.0191	0.0026	7.3399	0.0000	0.0140	0.0242
Vehicle Condition_Second Hand	0.0217	0.0027	8.1888	0.0000	0.0165	0.0269
Vehicle Condition_Zero Imported from Abroad	0.0043	0.0012	3.5715	0.0004	0.0019	0.0066
From_Manufacturer	-0.0010	0.0011	-0.9695	0.3323	-0.0031	0.0011
From_Owner	0.0227	0.0011	20.5823	0.0000	0.0205	0.0248
Traction_4x4	0.0003	0.0010	0.2533	0.8000	-0.0018	0.0023
Traction_Front-wheel-drive	0.0000	0.0010	0.0000	1.0000	-0.0000	0.0000
Traction_Rear-wheel-drive	0.0000	0.0010	0.0000	1.0000	-0.0000	0.0000
feature						
Year	1.66					
Kilometers (km)	1.53					
Engine Size (cc)	1.03					
Engine Power (hp)	1.64					
Fuel Tank Size (L)	1.46					
Transmission_Manual	1.81					
Transmission_Semi-Automatic	1.70					
Fuel Type_Electric	1.02					
Fuel Type_Gasoline	1.24					
Fuel Type_Hybrid	1.03					
Fuel Type_LPG and Gasoline	1.35					
Body Type_Cabrio	1.01					
Body Type_Coupe	1.03					
Body Type_Crossover	1.01					
Body Type_MPV	1.00					
Body Type_Other	1.06					
Body Type_SUV	1.00					
Body Type_Station Wagon	1.01					
Body Type_Van	1.02					
Vehicle Condition_New	6.26					
Vehicle Condition_Second Hand	6.52					
Vehicle Condition_Zero Imported from Abroad	1.33					
From_Manufacturer	1.06					
From_Owner	1.13					
Traction_4x4	1.00					
Traction_AWD (Electronic)	1.02					
Traction_Front-wheel-drive	6.92					

```
Best Ridge Regression Alpha: 719.6856730611514
Ridge Regression RMSE: 0.2968420852024057
Ridge Regression R2: 0.765737393607462
Ridge Regression Coefficients: [ 0.00000000e+00 3.81576342e-01 -6.41025184e-02 1.86753908e-02
2.13941349e-01 5.44023335e-02 -5.86752849e-02 -4.84040102e-03
-9.19195724e-03 2.28467966e-02 4.96185374e-03 1.08990239e-02
8.00013428e-03 1.43201616e-02 -4.18306745e-03 -1.11111723e-03
-8.31127224e-03 -3.26588486e-03 -7.85242875e-03 -7.16337793e-03
1.70374153e-02 1.97193957e-02 3.80222807e-03 -9.73679853e-04
2.21798118e-02 3.06426677e-04 1.69013925e-03 -4.05895151e-02
-5.37455671e-03]
Best Lasso Regression Alpha: 0.004291934260128779
Lasso Regression RMSE: 0.2975355331196055
Lasso Regression R2: 0.7646449681303382
Lasso Regression Coefficients: [ 0.00000000e+00 3.80807649e-01 -5.84902796e-02 1.37996842e-02
2.17225806e-01 4.86614965e-02 -5.23246281e-02 -0.00000000e+00
-5.06742321e-03 1.78597805e-02 3.40602420e-04 5.02335203e-03
4.90451879e-03 1.08471242e-02 -0.00000000e+00 -0.00000000e+00
-4.19977730e-03 -0.00000000e+00 -3.96257390e-03 -3.78639197e-03
-0.00000000e+00 9.87322212e-05 -0.00000000e+00 -0.00000000e+00
1.75218111e-02 0.00000000e+00 0.00000000e+00 -3.41723598e-02
0.00000000e+00]
```



- By adding one-hot encoding for “Make” and “Model” the Adj-R2 goes up to: 0.814
- Ridge and Lasso cannot get better values



Results: Ordinary least squares						
Model:	OLS	Adj. R-squared:	0.814			
Dependent Variable:	log_USD_Absolute	AIC:	14540.1237			
Date:	2024-06-05 22:43	BIC:	15405.9686			
No. Observations:	81647	Log-Likelihood:	-7177.1			
Df Model:	92	F-statistic:	3890.			
Df Residuals:	81554	Prob (F-statistic):	0.00			
R-squared:	0.814	Scale:	0.069883			
	Coef.	Std.Err.	t	P> t	[0.025	0.975]
const	9.7231	0.0009	10509.6556	0.0000	9.7213	9.7249
Year	0.3879	0.0013	303.9874	0.0000	0.3854	0.3904
Kilometers (km)	-0.0677	0.0012	-57.5783	0.0000	-0.0700	-0.0654
Engine Size (cc)	0.0120	0.0009	12.7261	0.0000	0.0102	0.0139
Engine Power (hp)	0.1728	0.0013	132.7936	0.0000	0.1703	0.1754
Fuel Tank Size (L)	0.0486	0.0012	40.0264	0.0000	0.0462	0.0509
Make_Alfa Romeo	-0.0096	0.0024	-3.9940	0.0001	-0.0143	-0.0049
Make_Apec	0.0003	0.0009	0.3544	0.7230	-0.0015	0.0022
Make_Audi	0.0409	0.0071	5.7818	0.0000	0.0271	0.0548
Make_BMC	0.0004	0.0010	0.4042	0.6860	-0.0015	0.0023
Make_BMW	0.0607	0.0004	15.3261	0.0000	0.0593	0.0621

Best Ridge Regression Alpha: 719.6856730011514
Ridge Regression RMSE: 0.26424664141717485
Ridge Regression R2: 0.8143629053558321
Ridge Regression Coefficients: [0.00000000e+00 1.72481322e-01 4.82599255e-02 -9.10070467e-03 4.20244507e-02 4.42176364e-04 6.92961083e-02

-1.06107026e-02]
Best Lasso Regression Alpha: 0.0006551285568595509
Lasso Regression RMSE: 0.2643050493729932
Lasso Regression R2: 0.8142808314155586
Lasso Regression Coefficients: [0.00000000e+00 3.870 1.73922843e-01 4.77361104e-02 -6.66544928e-03 0.00 4.71116402e-02 0.00000000e+00 7.40641102e-02 5.15

- Because of the large data set I had problems running these other models (Decision Tree / Random Forest / SVR, KNN, XGBoost) on the full set, so instead I ran them on a 10,000 sample of it, with these results:

```
print("\n")
```

Decision Tree Regressor Results:

RMSE: 0.1974222584026552

R2: 0.8994310389754823

Best Params: {'max_depth': 10}

Random Forest Regressor Results:

RMSE: 0.08709518531173507

R2: 0.9804268814216144

Best Params: {'max_depth': None, 'n_estimators': 50}

Support Vector Regressor Results:

RMSE: 0.24336364750158648

R2: 0.8471789624593722

Best Params: {'C': 10, 'epsilon': 0.1}

K-Nearest Neighbors Regressor Results:

RMSE: 0.25530594379525245

R2: 0.8318125509262988

Best Params: {'n_neighbors': 5}

XGBoost Regressor Results:

RMSE: 0.17057861111182665

R2: 0.9249205850051347

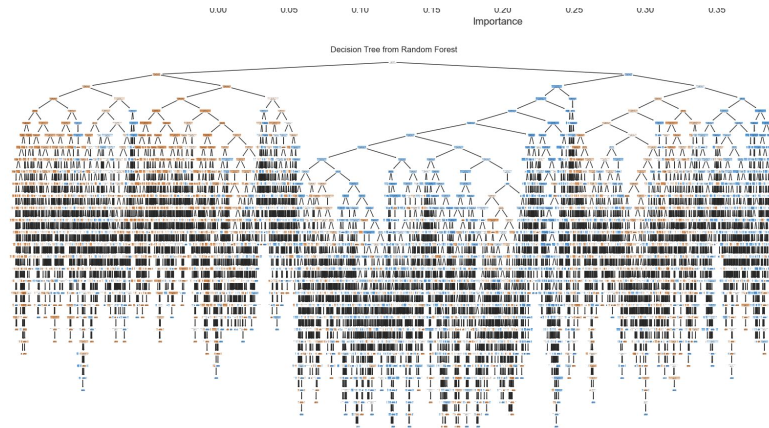
Best Params: {'learning_rate': 0.2, 'n_estimators': 50}

2.0 Data | Random Forest

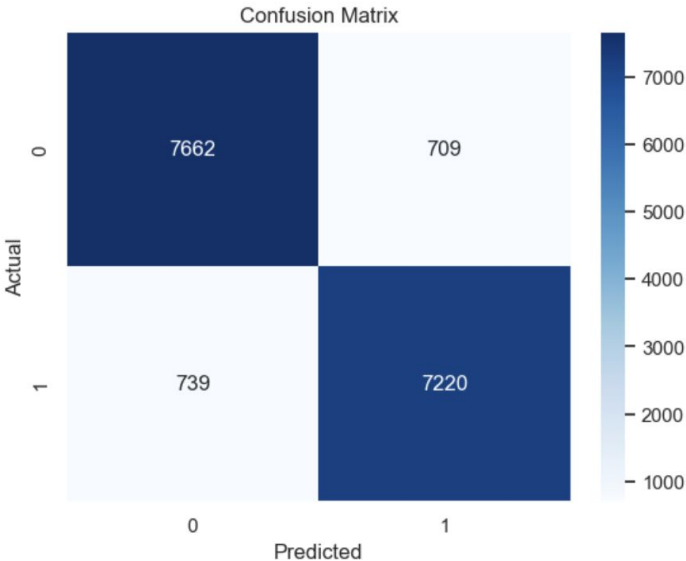
- The Random Forest result was very high so I ran that model on the full set, to see how it behaves

Random Forest Classifier Accuracy: 0.911328842620943
Random Forest Classifier Precision: 0.9105814100138732
Random Forest Classifier Recall: 0.907149139339113
Random Forest Classifier F1 Score: 0.9088620342396777

	precision	recall	f1-score	support
0	0.91	0.92	0.91	8371
1	0.91	0.91	0.91	7959
accuracy			0.91	16330
macro avg	0.91	0.91	0.91	16330
weighted avg	0.91	0.91	0.91	16330



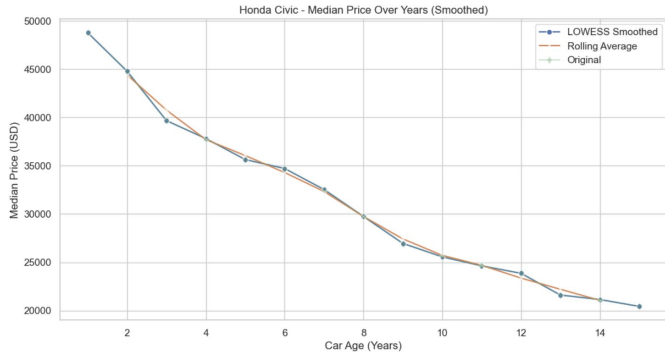
Fitting 3 folds for each of 216 candidates, totalling 648 fits



Calculate Percentage Lost every year

- For every Make-Model, calculate age in years;
- Calculate median mileage (km) and eliminate cars above 30% over median value (this is to eliminate high mileage/low value cars)
- Then group by car's age in years, calculate median USD Value
- Then from year to year - calculate Percentage Loss
- Smooth the graph
- Analyze results and pick the winners

Processing Honda Civic
Columns in median_prices for Honda Civic: Index(['Car Age (Years)', 'USD Absolute', 'Car Count', 'Smoothed Price', 'LOWESS Price', 'Percentage Loss'], dtype='object')



Honda Civic - Percentage Loss Over Years				
	Car Age (Years)	Percentage Loss	Car Count	USD Absolute
1	1	NaN	18	48759.65
2	2	-8.18	33	44770.58
3	3	-11.42	125	39658.37
4	4	-4.69	128	37799.38
5	5	-5.74	119	35630.57
6	6	-2.61	184	34701.07
7	7	-6.25	139	32532.26
8	8	-8.57	67	29743.78
9	9	-9.38	57	26955.30
10	10	-5.17	62	25561.06
11	11	-3.64	97	24631.57
12	12	-3.14	123	23856.99
13	13	-9.42	78	21609.16
14	14	-2.14	76	21145.95
15	15	-3.37	56	20433.35

Winners

By looking mostly at Percentage Loss but also at Kilometers (lower is better) and Car Count (higher is better) I decided to go with these 4 cars that add up to almost our \$100,000 budget

Summary statistics for Fiat Egea (6-9):					
Car Age (Years)	Percentage Loss	Car Count	USD Absolute	Kilometers (km)	
0	6	-0.74	61	20758.68	140000.00
1	7	-1.49	59	20448.85	152650.00
2	8	0.53	49	20557.29	138500.00
3	9	5.39	8	21664.93	143900.00

Summary statistics for Volkswagen Golf (4-7):					
Car Age (Years)	Percentage Loss	Car Count	USD Absolute	Kilometers (km)	
0	4	-23.53	30	32222.42	107000.00
1	5	7.69	56	34701.07	87000.00
2	6	-3.35	76	33539.21	105000.00
3	7	-3.93	129	32222.42	100000.00

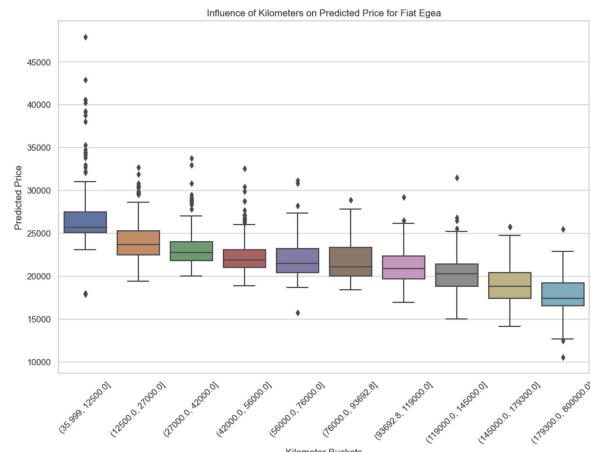
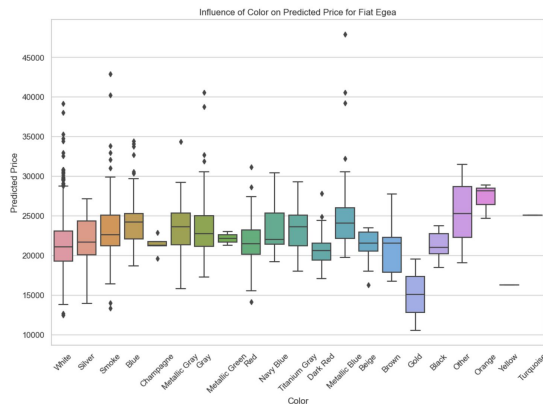
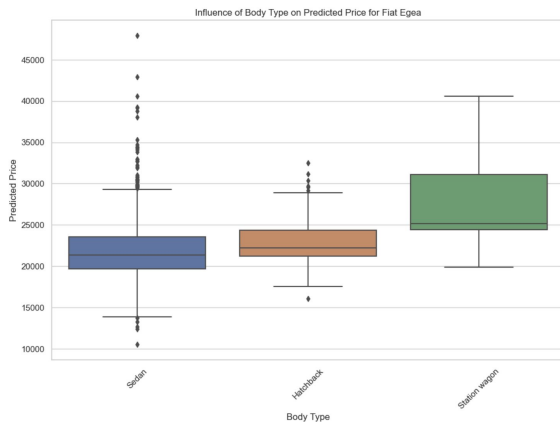
Summary statistics for Dacia Sandero (4-7):					
Car Age (Years)	Percentage Loss	Car Count	USD Absolute	Kilometers (km)	
0	4	-5.02	25	22217.96	51250.00
1	5	-2.80	42	21595.22	63000.00
2	6	4.02	39	22462.75	83000.00
3	7	-2.76	70	21843.09	90750.00

Summary statistics for Hyundai i20 (2-5):					
Car Age (Years)	Percentage Loss	Car Count	USD Absolute	Kilometers (km)	
0	2	-11.10	63	26304.65	30000.00
1	3	1.88	65	26800.38	38980.00
2	4	-4.60	30	25568.03	49156.00
3	5	-0.63	33	25406.14	58750.00

Winning Features

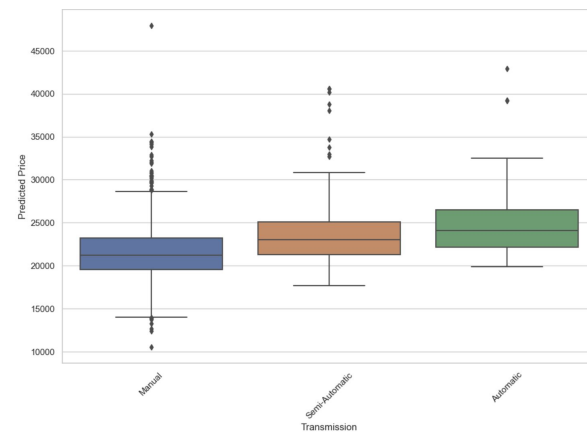
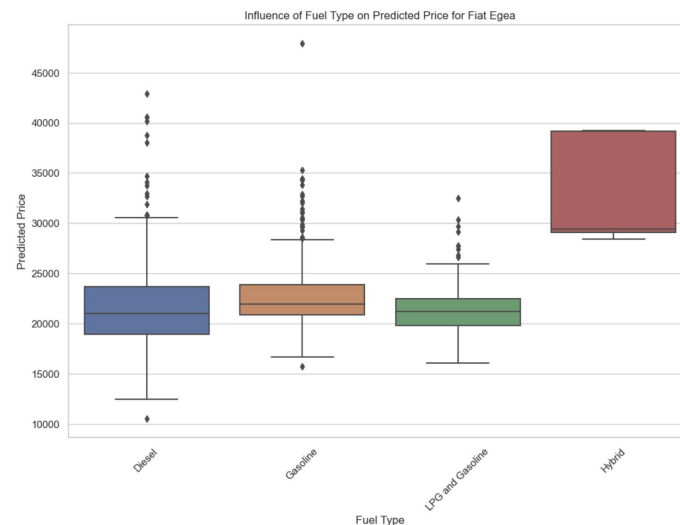
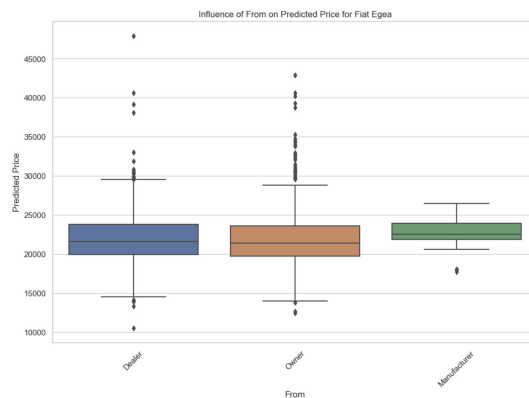
Now that we know what vehicles we are looking for, let's pick the other details where possible: color, body type, transmission etc

For this I used the model to predict the price; then from the model we extracted the influence of each variable on the Predicted Price.



So our ideal Fiat Egea would be:

- Station wagon
- Orange or Blue
- Anything under 150,000 km is acceptable but the lower the better
- Hybrid or Gasoline
- Automatic if possible
- Preferably from an Owner



Links

By running the model on this specific information, it runs Price Predictions and compares them with the listed price (USD Absolute) to find the best deals:

Sahibinden Dacia Sandero 0.9 TCe Stepway Easy-R 2020 Model Amasya 47.000 km Beyaz

Fatih Mh. Merkez, Afyonkarahisar



Fiyat

630.000 TL

Ekspertiz teklifleri

İlan No

25151648

İlan Tarihi

21 Nisan 2024

Marka

Dacia

Seri

Sandero

Model

0.9 TCe Stepway Easy-R

Yıl

2020

Kilometre

47.600 km

Vites Tipi

Yarı Otomatik

Yakıt Tipi

Benzin

Kasa Tipi

Hatchback/5

Renk

Beyaz

Motor Hacmi

898 cc

Motor Gücü

90 hp

Çekiş

Önden Çekiş

Araç Durumu

İkinci El

Ort. Yakıt Tüketimi

5,1 lt

1 / 15



Conclusion

By choosing 4 different cars as investment I believe I can offer these estimates for loss of value over 3 years:

- Optimistic - 4%
- Pessimistic - 10%

The used car prices are related to much larger macro-economic trends so this is just an estimate.

But considering the instability of the Turkish currency, I believe cars can be a way of sheltering from inflation.



Thank you!